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Control analysis of parallel DC-DC converters in a DC microgrid with constant power loads

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ABSTRACT

This paper proposes a general methodology for designing hierarchical control schemes for DC microgrids loaded by constant power loads. This paper addresses issues when interfacing sources with a wide voltage range allowing for any general choice for a power source. The control strategy consists of two levels. The lower level consists of droop-based primary controllers. This controller enables current-sharing among paralleled sources and also damps limit cycle oscillations due to constant power loads. The higher level is a secondary controller which compensates for voltage deviations due to primary controller and performs voltage control of the microgrid. It also maintains the current sharing obtained in the primary stage. In the proposed secondary control, high-speed communication links to the primary controllers is implemented. The stability conditions are explained using the equivalent circuit of converters. Using this approach, stability conditions can be derived for microgrids of an arbitrary size and converter topology. The proposed control scheme is shown to be scalable and robust. The results obtained for the three basic DC-DC converter configurations are compared based on a microgrid with a parallel configuration of buck-boost converters. Simulations and experimental results are presented to verify the validity of the proposed control schemes.

1. Introduction

Recent energy concerns include power systems reliability, availability and resilience, increase in energy consumed beyond of existing transmission capacity [1,2]. With increasing penetration of distributed generation (DG), microgrids have come to coexist with traditional power system architectures [3]. The DC microgrid test bed implemented by the city college of New York (CCNY) [4] is an example of a microgrid coexisting with the traditional power grid. Analysis in [5] showed that microgrids with diverse sources and a redundant architecture can achieve high availability suitable for critical applications, such as data centers [6] with air-conditioning systems [7]. Compared to conventional grid, microgrids seem to be robust for critical infrastructures in the event of natural disasters [8]. Although more development work has focused on AC microgrids, recently DC microgrids are also receiving renewed attention [9]. DC microgrids do not have frequency or reactive power related issues and, hence, analysis and design of control loops may be simpler [10]. In addition, a number of sources like solar panels, fuel cells, energy storage and present-day electronic loads are inherently DC [11]. In case of power supply for telecom networks, availability is improved by supplying power with a DC distribution architecture [12].

One of the main challenges for microgrids is that constant power loads (CPLs) may constitute a major proportion of the load [13,14]. The presence of CPLs may cause limit cycle oscillations in the microgrid voltage and currents due to their negative incremental resistance [15]. An important preference to improve microgrids availability is to use distributed and autonomous control of the constituent converters [16]. This issue is addressed in [17] in which a general framework of a hierarchical control scheme for both grid-connected as well as islanded microgrids was introduced. Based on the framework in [17], hierarchical control scheme is proposed in [18–20] for a multi-bus DC microgrid with low bandwidth communication based secondary control scheme. In the method proposed in [21], system damping is reduced by lumping the loads to a single bus. The methods proposed in [18–21] do not seem to address the issues due to CPLs in the microgrids. This is because the methods proposed in [18–21] assume that microgrid loads are resistive and not CPLs. The review paper on hierarchical control [22] states that, the objectives of achieving current sharing accuracy and voltage restoration by using secondary control are complementary.

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Thus, the objective of accurate current sharing in addition to voltage regulation is yet to be achieved. Some recent work in [23,24] try to address this issue by using non-linear droop characteristics. But some non-linear droop characteristics may lead to unstable equilibrium points in the presence of CPLs [24]. Nevertheless, using the methods proposed in [23,24] leads to steady state voltage deviation under high load conditions. However, secondary control proposed in this paper achieves zero steady-state error in the microgrid voltage even under heavy loads

A significant sub-domain of research on hierarchical control involves interfacing renewable energy sources (RESs) and energy storage systems (ESSs) to the DC microgrid. In this related area, the function of secondary controller is mainly on efficient utilization of RES and ESS [25–27]. In this context in [25,26] ESS, solar, wind turbine generators are interfaced to a microgrid. The method proposed in [27] uses batteries and ultracapacitors interfaced to a DC microgrid using converters. In the methods proposed in [25–27], microgrid voltage is allowed to vary and is not tightly regulated. The method proposed in [28,29] uses additional ESSs with adjustable droops. In [30], a control scheme for minimizing the losses of the DC-DC converters in the microgrid is implemented. However, the main objective of secondary control in [28–30] seems to be that of balancing the state of charge of the ESS. In [31], the goal of secondary controller is the economic dispatch of sources connected to the microgrid. In [32] the objective of secondary control is that of efficient charging of ESS. Even if some of the works like [33,34] consider microgrid loads as CPLs, the source interfacing converters are of buck or boost topology. Detailed quantitative analysis in [35] showed that microgrids with diverse sources can achieve high availability. In order to maximize the range of source voltages that could be interfaced, converters capable of providing wider output voltage ranges may be required [36]. Further, microgrid stability conditions derived in [18–22,25–34] only pertain to the respective systems considered and it may not be generalized. In this paper, a single bus DC microgrid is considered which is consistent with the existing approach [37]. In addition, a single bus microgrid also minimizes the effect of system damping. In a recent work [38], connecting lines, filters and resistive loads have been considered, which may partly aid in damping CPL oscillations. This paper does not consider such damping elements and hence the proposed control scheme entirely damps the limit cycle oscillations. In addition, the system efficiency is higher than that in [38] because there is no power dissipation in added passive elements. In the later part of this paper, connecting lines have been considered in order to show that the proposed control scheme works well in the presence of other circuit elements.

This paper presents following novel contributions.

1. Hierarchical control schemes are proposed for DC microgrids with CPLs instead of linear loads.
2. The primary controller performs dual functions of current sharing as well as damping limit cycle oscillations due to CPLs.
3. The secondary controller performs voltage regulation. The integral control gains of the secondary controller are designed such that current sharing obtained using primary control is maintained. Thus, this paper addresses complementary issues listed in [22].
4. Secondary control is implemented using a controller area network (CAN) communication protocol [20]. This ensures better accuracy in current sharing and avoids instability due to racing condition in the event of multiple controllers trying to regulate microgrid voltage [39].
5. Stability conditions are derived using a computationally simpler approach based on the converter equivalent circuit model. The proposed approach is scalable in terms of microgrid size and number of power sources.
6. This paper addresses issues when interfacing sources with a wide voltage range allowing for any general choice for a power source. The basic DC-DC converter configurations that satisfy this condition is the DC-DC buck-boost converter. Higher order converters may have additional advantages. But they contain additional passive components which may increase converters failure rates [40]. Hence, each stage of controller design process and stability analysis is explained by considering a DC microgrid containing parallel buck-boost converters. This approach is shown to be scalable to microgrids of an arbitrary topology, converter configuration and in the presence of additional microgrid elements. To cover a wide range of source converter topologies, results obtained for the 3 basic DC-DC converter topologies are also compared.

Each of the contributions mentioned in the previous paragraph are improvements over the work presented in the current literature. In [41], microgrid loads are assumed to be constant current loads. In this paper, microgrid loads are assumed as CPLs which is more realistic. The active damping method proposed in [42] only damps the limit cycle oscillations, whereas primary controller proposed in this paper performs current sharing as well as damping oscillations. In [43], load current sharing is performed with the help of centralized communication scheme and hence it is not autonomous. Since droop scheme is used for current sharing in this paper, the primary controller is fully autonomous which takes advantage in order to plug and play various microgrid sources. In [44], a low bandwidth based communication scheme is used for current sharing among converters. If primary control relies on communication links, limit cycle oscillations in currents and voltages are observed in the occasion of failure of such links. Local autonomous primary controller proposed in this paper, allows the microgrid to still operate, although, possibly, in a suboptimal state, even when communication links fail. Under light loaded conditions, microgrid voltage level may still be acceptable. Methods to perform voltage regulation of DC microgrid locally have been proposed in literature in [45]. However, this could eventually lead to instability due to differences in sensor parameters [39]. In this paper, secondary control is implemented using a controller area network (CAN) communication protocol [20] which performs voltage control of the microgrid from a single location. Additionally, in a recent work [23] while performing stability analysis, converters are modeled as voltage sources in series with a resistance, which may lead to model inaccuracies. Such simple models do not provide any information regarding the effects of converter parameters on microgrid stability. In this paper, a detailed model of source converters is considered and no state variables are neglected. Even though [46] mentions that buck-boost converters are used as source interfacing converters, it assumes that CPLs fed by such converters are inherently stable. Further, no information regarding the converter model seems to be available in [46].

This paper is organized as follows. In Section 2, a general methodology for designing the primary and secondary controllers for a DC microgrid containing non-isolated DC-DC converters is presented. In Section 3, equivalent circuits of the converters are used to obtain stability conditions. Further, stability of the proposed control scheme is also studied. Simulation and experimental results are presented in Section 4 to verify the proposed control strategies. In each of the sections, DC-DC buck-boost topology is used to describe the results derived and the results obtained for the three basic DC-DC converter topologies are compared.

2. Controller design and analysis

Consider a system of m parallel-connected source converters supplying power to r CPLs [47] as shown in Fig. 1. The source converters are assumed to be operating in continuous conduction mode (CCM). The average state equations for such a system are

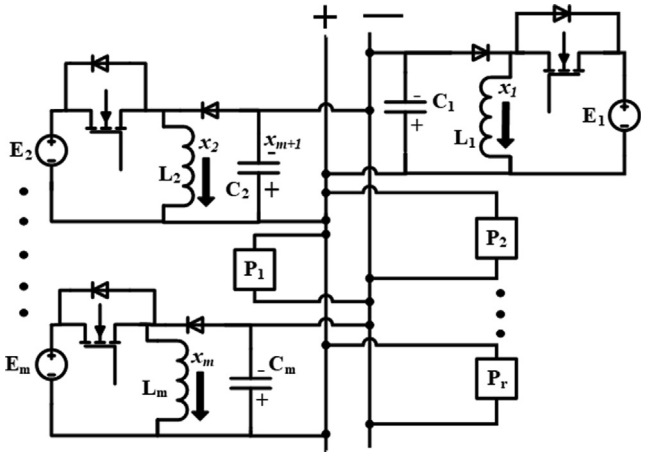


Fig. 1. A system of m parallel buck-boost converters supplying r CPLs.

$$\mathbf{f}(\mathbf{x}, t) = \begin{cases} f_1(\mathbf{x}, \mathbf{E}, P, \mathbf{d}) \\ f_2(\mathbf{x}, \mathbf{E}, P, \mathbf{d}) \\ \vdots \\ f_n(\mathbf{x}, \mathbf{E}, P, \mathbf{d}) \end{cases}$$

$$P = P_1 + P_2. \ldots..+P_r$$

$$\mathbf{d} = [d_1, d_2, \ldots d_m]^T, \mathbf{E} = [E_1, E_2, \ldots E_m]^T, \mathbf{x} = [x_1, x_2, \ldots x_n]^T \quad (1)$$

where $\mathbf{x}$ is the state vector with components given by the inductor currents and capacitor voltages. It is assumed that there are n state variables in the system with $n = m + 1$. The instantaneous duty cycle and source voltage of the ith converter are given by d_i and E_i, respectively. To ensure stable operation of source converters, consider the vector of duty cycles, $\mathbf{d}$ in (1) to involve state variables feedback. Hence, let

$$\mathbf{d} = \mathbf{g}(\mathbf{x}) \quad (2)$$

From a control standpoint, the individual components of $\mathbf{g(x)}$, namely $g_1, g_2 \ldots g_m$ may be any linear or non-linear functions of $\mathbf{x}$. However, from the system and implementation standpoint, there are several constraints to be taken into account while deriving $\mathbf{g(x)}$. The aim of this section is to take into account each of these constraints and arrive at a feasible option.

Constraint 1: The chosen state feedback maintains the asymptotic stability of the corresponding equilibrium point, $\mathbf{X_{eq}}$. The stability of $\mathbf{X_{eq}}$ is estimated by linearizing the equations about the equilibrium point [48]. As an example, consider the system shown in Fig. 1. The average state space equation for the system is

$$\mathbf{f}(\mathbf{x},t) = \begin{cases} L_j \frac{dxj}{dt} = d_j E_j - d_j' x_{m+1} \\ (C_1 + \ldots + C_m)\frac{dx_{m+1}}{dt} = \sum_{j=1}^{m} (d_j' x_j) - \frac{P}{x_{m+1}} \end{cases}$$

$$d_j' = 1 - d_j, j = 1, 2, \ldots, m$$

$$\mathbf{x} = [x_1, \ldots x_m, x_{m+1}]^T = [i_{L1}, \ldots, i_{Lm}, v_C]^T \quad (3)$$

where $d_1, d_2, \ldots, d_m$ refer to the instantaneous duty cycles of the m parallel converters.*Constraint 2:* The duty ratio function of one converter cannot include the feedback of inductor current of another converter. This is important to ensure the plug and play operation of the microgrid sources [40]. Hence

$$d_j = g_j(x_j, x_{m+1}), j = 1, 2. ., m \quad (4)$$

Constraint 3: The primary controller is fully autonomous [16] This means that primary controller of a converter can only include a feedback signal of its own inductor current. Hence let's assume that

$$d_j = g_j(x_j), j = 1, 2. ., m \quad (5)$$

so that the control laws are symmetric. Calculating the system jacobian, $\mathbf{A}$ we obtain

$$A_{(m+1)\times(m+1)} = \begin{bmatrix} \frac{(E_1 + x_{m+1})}{L_1}\frac{\partial g_1}{\partial x_1} & \ldots & 0 & \frac{1}{L_1}(g_1 - 1) \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & \frac{(E_m + x_{m+1})}{L_m}\frac{\partial g_m}{\partial x_m} & \frac{1}{L_m}(g_m - 1) \\ \frac{1}{\sum_{j=1}^{m} C_j}\left(1 - g_1 - x_1\frac{\partial g_1}{\partial x_1}\right) & \cdots & \frac{1}{\sum_{j=1}^{m} C_j}\left(1 - g_m - x_m\frac{\partial g_m}{\partial x_m}\right) & \frac{P}{\sum_{j=1}^{m} C_j x_{m+1}^2} \end{bmatrix} \quad (6)$$

It can be observed from (6) that in order for all the $m + 1$ eigenvalues to lie on the left half of the s plane (LHP), one of the necessary conditions is that trace of $\mathbf{A}$ must be negative. Hence,

$$\frac{\partial g_j}{\partial x_j} < 0 \quad (7)$$

From an implementation standpoint, the simplest means to ensure that (7) is satisfied is by providing a proportional negative feedback of the inductor currents of the respective converters.

Constraint 4: Duty ratio constraint. Since $0 < d_j < 1$ in (5), the following control laws can be used for the primary controller in (5).

$$d_j = \frac{k_{OL,j}E_j - R_{dj}x_j}{E_j}$$

$$\frac{k_{OL,j}E_j}{1 - k_{OL,j}} = V_{nl}, 0 < k_{OL,j} < 1 \quad (8)$$

where V_{nl} refers to the no load voltage of the DC microgrid, $k_{OL,j}$ and R_{dj} are the open-loop duty ratio and virtual droop resistance of the jth converter respectively. The control inputs of the jth converter given by (8) are conveniently referred to as the primary control inputs $d_{j,pri}$. The equilibrium point obtained using the control inputs (8) is given by $\mathbf{X_{eq,pri}}$. The individual components of $\mathbf{X_{eq,pri}}$ are given by $[X_{1eq}, X_{2eq} \ldots X_{(m+1)eq}]^T$. Strict droop controllers enable current sharing at the cost of poor voltage regulation. This can be remedied by adding secondary control inputs containing integral control terms [17]. The overall control inputs are then given by

$$d_j = \frac{k_{OL,j}E_j - x_j R_{dj} + k_{i,j}\int (V_{nl} - x_{m+1})dt}{E_j} \quad (9)$$

where $k_{i,j}$ is the integral control gain of the jth converter. The equilibrium point obtained using the control inputs (9) is given by $\mathbf{X_{eq,sec}}$. The individual components of $\mathbf{X_{eq,sec}}$ are given by $[X_{1eq,sec}, X_{2eq,sec} \ldots X_{(m+1)eq,sec}]^T$. To obtain the expressions for equilibrium points, current sharing ratios and integral control gains in terms of other state variables, consider the case where $m = 2$ without loss of generality. For a multi-converter system such as the one in Fig. 1, the variation of microgrid voltage with load power is of considerable interest. Consider the system equations (3). At $\mathbf{X_{eq,pri}}$, the LHS of (3) is zero. To obtain X_{1eq} and X_{2eq} in terms of X_{3eq}, we do the following. The control input (8) is substituted in the first equation in (3). The variables x_1, x_2 and x_3 are replaced by X_{1eq}, X_{2eq} and X_{3eq} respectively. Writing X_{1eq} and X_{2eq} in terms of X_{3eq} in the first equation in (3), we can obtain

$$X_{jeq} = \frac{k_{OL,j}E_j^2 + E_j X_{3eq}(k_{OL,j} - 1)}{R_{dj}(E_j + X_{3eq})}, j = 1, 2 \quad (10)$$

Writing X_{1eq}, d_1', X_{2eq}, d_2' in terms of X_{3eq} in the second equation in (3) and ordering terms according to powers of X_{3eq}, we can obtain (11).The equilibrium point, $\mathbf{X_{eq,pri}}$ can be obtained in terms of R_{d1}, R_{d2}, E_1, E_2, $k_{OL,1}$, $k_{OL,2}$ and P by solving the following quartic equation.

$$f_{pri}(X_{3eq}): a_4 X_{3eq}^4 + a_3 X_{3eq}^3 + a_2 X_{3eq}^2 + a_1 X_{3eq} + a_0 = 0 \quad (11)$$

where

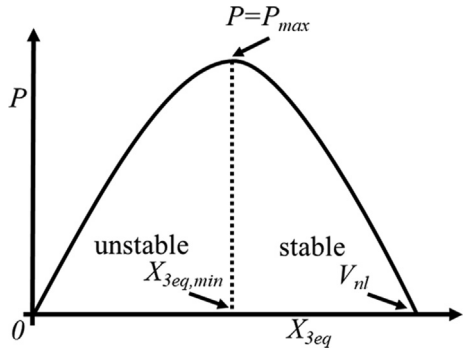


Fig. 2. X_{3eq} Vs P characteristics for parallel buck-boost converters in Fig. 1 with $m = 2$ [25].

$$\begin{aligned} a_4 &= -E_1^2(1-k_{OL,1})R_{d2} - E_2^2(1-k_{OL,2})R_{d1} - PR_{d1}R_{d2} \\ a_3 &= E_1^3 k_{OL,1}R_{d2} + E_2^3 k_{OL,2}R_{d1} - 2R_{d1}E_2^2E_1(1-k_{OL,2}) \\ &\quad - 2R_{d2}E_1^2E_2(1-k_{OL,1}) \\ &\quad - 2PR_{d1}R_{d2}(E_1+E_2) \\ a_2 &= 2E_1^3E_2k_{OL,1}R_{d2} + 2E_2^3E_1k_{OL,2}R_{d1} - E_1^2(1-k_{OL,1})E_2^2R_{d2} \\ &\quad - E_2^2(1-k_{OL,2})E_1^2R_{d1} - 4PR_{d1}R_{d2}E_1E_2 - PR_{d1}R_{d2}(E_1^2+E_2^2) \\ a_1 &= E_1^3E_2^2k_{OL,1}R_{d2} + E_2^3E_1^2k_{OL,2}R_{d1} - 2PR_{d1}R_{d2}E_1E_2(E_1+E_2) \\ a_0 &= -E_1^2E_2^2PR_{d1}R_{d2} \end{aligned} \tag{12}$$

The significance of (11) is as follows. For practical values of E_1, E_2, R_{d1}, R_{d2}, $k_{OL,1}$ and $k_{OL,2}$, X_{3eq} Vs P characteristic is indicated in Fig. 2. Of the four roots from (11), two of them are always complex. The nature of the other two roots is as follows. When $P = 0$, one of the other two roots is $X_{3eq} = V_{nl}$ and the remaining other is $X_{3eq} = 0$. As P is increased, both the real roots start moving towards each other. At $P = P_{max}$, both of these roots equal to $X_{3eq,\min}$. Beyond P_{max}, there are no real roots for the equation (11). Thus, the locus of real root which starts at $X_{3eq} = V_{nl}$ for $P = 0$ and moves left till it reaches $P = P_{max}$ represents the locus of microgrid voltage. The characteristic obtained in Fig. 2 is similar to that obtained for parallel buck converters in [25,59], however, there are differences. In the case of parallel buck converters, $X_{3eq,\min}$ is a constant and is given by

$$X_{3eq,\min} = \frac{V_{nl}}{2} \tag{13}$$

In case of parallel buck-boost converters,

$$0 < X_{3eq,\min} \leqslant \frac{V_{nl}}{2} \tag{14}$$

For stability considerations, the parallel buck-boost operation can be restricted to lie in the region

$$\frac{V_{nl}}{2} \leqslant X_{3eq} \leqslant V_{nl} \tag{15}$$

The primary control inputs (8) enable current sharing according to

$$q_{I\,share} = \frac{X_{1eq}}{X_{2eq}} \tag{16}$$

The equilibrium point, $\mathbf{X_{eq,pri}}$ obtained in terms of X_{3eq} is given by

$$\mathbf{X_{eq,pri}} = \begin{bmatrix} X_{1eq} \\ X_{2eq} \\ X_{3eq} \end{bmatrix} = \begin{bmatrix} \frac{E_1(1-k_{OL,1})(V_{nl}-X_{3eq})}{R_{d1}(X_{3eq}+E_1)} \\ \frac{E_2(1-k_{OL,2})(V_{nl}-X_{3eq})}{R_{d2}(X_{3eq}+E_2)} \\ X_{3eq} \end{bmatrix} \tag{17}$$

Table 1 gives the results regarding primary equilibrium point, $\mathbf{X_{eq,pri}}$ for three basic configurations.

For ideal microgrid operation, it is desired that current sharing obtained with the primary control stage is maintained after the addition of the secondary controller. In the case of simple converter topologies, this is possible only if the converter inductor is either at the input or the output of the converter. The term q_{Ishare} is also used to represent the ratio of inductor currents in the secondary control stage similar to (16). In the case of parallel buck-boost converters, the equilibrium point obtained using the control inputs (9) is given by

Table 1
Primary control results for the 3 basic topologies [59,60].

Quantity	Buck	Boost	Buck-boost
X_{3eq} at $P = 0$	$V_{nl} = k_{OL,1}E_1$	$V_{nl} = \frac{E_1}{(1-k_{OL,1})}$	$V_{nl} = \frac{E_1 k_{OL,1}}{(1-k_{OL,1})}$
X_{3eq} at $P = P_{max}$	$\frac{V_{nl}}{2}$	E_1 if $E_1 > E_2$ E_2 if $E_1 < E_2$	$0 < X_{3eq,min} \leqslant \frac{V_{nl}}{2}$
q_{Ishare}	$\frac{R_{d2}}{R_{d1}}$	$\frac{E_1^2R_{d2}}{E_2^2R_{d1}}$	$\frac{(1-k_{OL,1})^2R_{d2}(E_2+X_{3eq})k_{OL,2}}{(1-k_{OL,2})^2R_{d1}(E_1+X_{3eq})k_{OL,1}}$

$$\mathbf{X_{eq,sec}} = \begin{bmatrix} X_{1eq,sec} \\ X_{2eq,sec} \\ X_{3eq,sec} \end{bmatrix} = \begin{bmatrix} \frac{k_{i,1}PR_{d2}}{V_{nl}(k_{i,1}R_{d2}(1-k_{OL,1})+k_{i,2}R_{d1}(1-k_{OL,2}))} \\ \frac{k_{i,2}PR_{d1}}{V_{nl}(k_{i,1}R_{d2}(1-k_{OL,1})+k_{i,2}R_{d1}(1-k_{OL,2}))} \\ V_{nl} \end{bmatrix} \tag{18}$$

We need to calculate the input currents of the top and bottom converters in the primary stage. Consider Fig. 3 which contains the equivalent circuit of the system in Fig. 1 with $m = 2$ and hierarchical control scheme of (8) and (9) implemented. The vector of input currents in the primary stage is given by $\mathbf{I_{in,pri}}$. The individual components of $\mathbf{I_{in,pri}}$ are given by $[I_{in1,pri}I_{in2,pri}]^T$. The transformer voltages in Fig. 3 can be calculated after substituting the equilibrium point (17) in the italicized expressions in Fig. 3. The current through the circuit elements can be calculated after equating the input and output power of the transformer. Thus, the input current of the top converter, $I_{in1,pri}$ can be calculated by simplifying the expression (19) after substituting for X_{1eq} from (17)

$$E_1 I_{in1,pri} = X_{1eq}\left(\frac{E_1 X_{3eq}}{X_{3eq}+E_1}\right) \tag{19}$$

Calculating $\mathbf{I_{in,pri}}$, we get

$$\mathbf{I_{in,pri}} = \begin{bmatrix} I_{in1,pri} \\ I_{in2,pri} \end{bmatrix} = \left[\frac{E_j(1-k_{OL,j})(V_{nl}-X_{3eq})X_{3eq}}{R_{dj}(E_j+X_{3eq})^2}\right]_{2X1}, j = 1, 2 \tag{20}$$

Consider the ratio of the input currents of the converters similar to (16). Rewriting $X_{3eq} = V_{nl}$-V_{dr} (V_{dr} is the droop voltage), substituting for V_{nl} from (8) and simplifying

$$\frac{I_{in1,pri}}{I_{in2,pri}} = \frac{E_1(1-k_{OL,1})R_{d2}\left(\frac{E_2}{(1-k_{OL,2})}-V_{dr}\right)^2}{E_2(1-k_{OL,2})R_{d1}\left(\frac{E_1}{(1-k_{OL,1})}-V_{dr}\right)^2} \tag{21}$$

Since $k_{OL,2} < 1$ and $k_{OL,1} < 1$ and considering

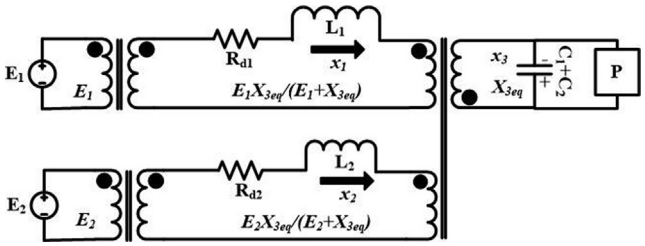


Fig. 3. Equivalent circuit of system in Fig. 1 ($m = 2$) with control scheme (8), (9) implemented [49]. The italicized values at the bottom represent transformer voltages in circuits.

$$\frac{E_j}{(1-k_{OL,j})} > V_{nl} > V_{dr}, j = 1, 2 \tag{22}$$

then, from (22) and (15), V_{dr} can be neglected in (21) and the following approximation can be made.

$$\frac{I_{in1,pri}}{I_{in2,pri}} \approx \frac{E_2(1-k_{OL,1})^3 R_{d2}}{E_1(1-k_{OL,2})^3 R_{d1}} \tag{23}$$

The vector of input currents in the secondary stage is given by $\mathbf{I_{in,sec}}$. The individual components of $\mathbf{I_{in,sec}}$ are given by $[I_{in1,sec}, I_{in2,sec}]^T$. For calculating $\mathbf{I_{in,sec}}$, we follow the same procedure that we did in order to calculate $\mathbf{I_{in,pri}}$ but in Fig. 3, we use values from (18) instead of (17). Calculating $\mathbf{I_{in,sec}}$, from (18) it yields

$$\mathbf{I_{in,sec}} = \begin{bmatrix} I_{in1,sec} \\ I_{in2,sec} \end{bmatrix} = \begin{bmatrix} \frac{k_{i,1} P R_{d2} k_{OL,1}}{V_{nl}(k_{i,1}R_{d2}(1-k_{OL,1}) + k_{i,2}R_{d1}(1-k_{OL,2}))} \\ \frac{k_{i,2} P R_{d1} k_{OL,2}}{V_{nl}(k_{i,1}R_{d2}(1-k_{OL,1}) + k_{i,2}R_{d1}(1-k_{OL,2}))} \end{bmatrix} \tag{24}$$

Equating (23) and (24) to equalize the shared current (at the converter input) in the primary and secondary stages we obtain

$$\begin{aligned} k_{i,1} &= k_i k_{OL,2} E_2 (1-k_{OL,1})^3 \\ k_{i,2} &= k_i k_{OL,1} E_1 (1-k_{OL,2})^3 \end{aligned} \tag{25}$$

where k_i is the integral control constant which is common for all the converters. Table 2 provides the results regarding the secondary controller for the 3 basic converter configurations. A control diagram for an m parallel converter configuration is given in Fig. 4.

3. Stability conditions

3.1. Circuit analysis of stability conditions

The system's characteristic equation calculated at $\mathbf{X_{eq,pri}}$ is given by

$$\det(s\mathbf{I} - \mathbf{A}) = s^n + a_{n-1}s^{n-1} + \ldots\ldots\ldots + a_1 s + a_0 \tag{26}$$

where the coefficients a_{n-1}, $\ldots a_1$, a_0 are real numbers. Among the necessary conditions to ensure local asymptotic stability, those that can be obtained directly using the $\mathbf{A}$ matrix (without calculating its eigenvalues) are given by the following equations [48].

$$a_{n-1} > 0 \Rightarrow trace(\mathbf{A}) < 0 \Rightarrow \sum_{i=1}^{n} \mathbf{A}_{i,i} < 0 \tag{27}$$

$$a_0 > 0 \Rightarrow \begin{cases} \det(\mathbf{A}) < 0 & \text{if } n \text{ is odd} \\ \det(\mathbf{A}) > 0 & \text{if } n \text{ is even} \end{cases} \tag{28}$$

where $\mathbf{A}_{i,i}$ is the ith diagonal element of $\mathbf{A}$. To explain the condition given by (27), consider the following. The primary controllers are designed so a virtual resistance is added in series with each converter inductor as shown in Fig. 3. Thus,

$$|\mathbf{A}_{i,i}| = \left|\frac{\partial^2 x_i}{\partial t \partial x_i}\right| = \begin{cases} \frac{R_{di}}{L_i} = \left|\frac{1}{\tau_{i,R-L}}\right|, & 1 \leqslant i \leqslant m \\ \frac{P}{\left(\sum_{q=1}^{m} C_q\right) X_{\mu g,eq}^2} = \left|\frac{1}{\tau_{R-C}}\right|, & i = n \end{cases} \tag{29}$$

where $X_{\mu g,eq}$ is the equilibrium value of the microgrid voltage, $\tau_{i,R\text{-}L}$ and $\tau_{R\text{-}C}$ are the time constants of the ith R-L circuit and the output R-C circuit respectively. Using (29), the condition given by (27) can be written as

$$\sum_{i=1}^{m} \left|\frac{1}{\tau_{i,R-L}}\right| > \left|\frac{1}{\tau_{R-C}}\right| \tag{30}$$

If the CPL and droop resistance are located on different sides of transformer as in Fig. 3, they have to be referred to the same side of the transformer by multiplying by the appropriate voltage transformation ratios. The physical significance of the condition (30) is that the rate of rise of current in the input R-L circuits should be greater than the rate of rise of current in the output R-C circuit. Calculating Det($\mathbf{A}$) in (28) results in a series circuit comprising of an equivalent voltage source, source and load impedances of the microgrid. Evaluating the stability condition for the circuit described above yields the Middlebrook criterion, which is used for verifying stability of converters with CPLs [13,50]. The Middlebrook criterion applied to microgrids, can be restated as follows [47,51]. Microgrids power distribution architecture consist of cascaded converters. In microgrids, the equilibrium point stability is determined by the ratio of Z_s/Z_l where Z_s is the output impedance of the source converters and Z_l is the input impedance of the load converters. Equilibrium point stability can be ensured if Z_s/Z_l does not encircle $-1 + j0$ point. In a conservative sense, stability can be ensured if $|Z_s| < |Z_l|$ at all frequencies [47]. It has been mentioned in [40] that in the case of a DC microgrid, the source and load impedances consist only of resistive terms. Observing from the microgrid terminal, equivalent source and load resistances can be calculated and the stability condition given by (28) can be stated as

$$|R_{source}| < |R_{CPL}| \tag{31}$$

where R_{source} is the equivalent source resistance and R_{CPL} is the linearized resistance of CPL.

The stability criteria listed in (27) and (28) are examined for system in Fig. 1. Consider the characteristic polynomial for the $\mathbf{A}$ matrix given by (26). Following the steps explained in the previous paragraph and calculating the condition given by (27), we obtain

$$\sum_{q=1}^{m} \frac{R_{dq}}{L_q}\left(1 + \frac{X_{\mu g,eq}}{E_q}\right) > \frac{P}{\left(\sum_{q=1}^{m} C_q\right) X_{\mu g,eq}^2} \tag{32}$$

where R_{dq}, L_q, E_q and C_q are the droop resistance, inductance, source voltage and capacitance of the qth converter and $X_{\mu g,eq}$ is the microgrid equilibrium voltage. To obtain the condition corresponding to (28) consider the equivalent circuit of the parallel buck-boost configuration given in Fig. 3. Observing from the microgrid terminal and making use of the expression (31), we obtain

Table 2
Secondary control results for the 3 basic topologies [59,60].

Quantity	Buck	Boost	Buck-boost
Integral control gain	$k_{i,1} = k_{i,2} = k_i$	$k_{i,1} = k_i E_1^2$ $k_{i,2} = k_i E_2^2$	$k_{i,1} = k_i k_{OL,2} E_2 (1-k_{OL,1})^3$ $k_{i,2} = k_i k_{OL,1} E_1 (1-k_{OL,2})^3$
q_{Ishare}	$\frac{R_{d2}}{R_{d1}}$	$\frac{E_1^2 R_{d2}}{E_2^2 R_{d1}}$	$\frac{k_{OL,2} E_2 (1-k_{OL,1})^3 R_{d2}}{k_{OL,1} E_1 (1-k_{OL,2})^3 R_{d1}}$

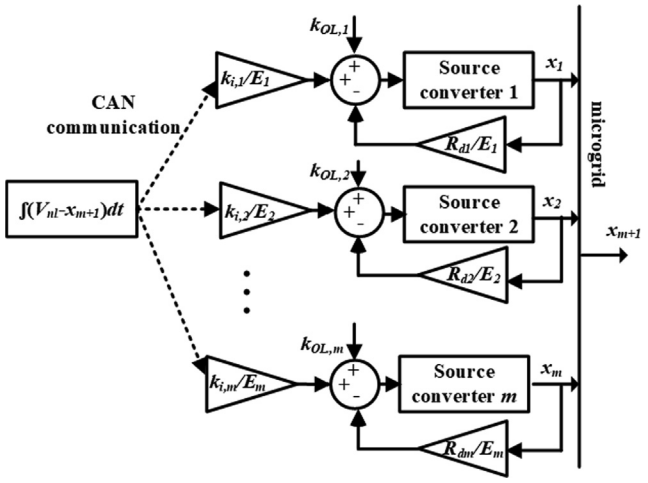


Fig. 4. Control architecture for the DC microgrid.

Table 3
Stability results for the buck and boost topologies [59,60].

Condition	Buck	Boost
$a_2 > 0$	$\sum_{q=1}^{m} \frac{R_{dq}}{L_q} > \frac{P}{\left(\sum_{q=1}^{m} C_q\right) X_{\mu g,eq}^2}$	$X_{\mu g,eq}\left[\sum_{q=1}^{m} \frac{R_{dq}}{L_q E_q}\right] > \frac{P}{\left(\sum_{q=1}^{m} C_q\right) X_{\mu g,eq}^2}$
$a_0 > 0$	$\sum_{q=1}^{m} \frac{1}{R_{dq}} > \frac{P}{X_{\mu g,eq}^2}$	$\frac{1}{X_{\mu g,eq}^2} \sum_{q=1}^{m}\left[\frac{E_q^2}{R_{dq}}\right] > \frac{P}{X_{\mu g,eq}^2}$

$$\sum_{q=1}^{m}\left(\frac{1}{R_{dq}}\left[\frac{E_q}{E_q + X_{\mu g,\,eq}}\right]^2\right) > \frac{P}{X_{\mu g,\,eq}^2} \tag{33}$$

Table 3 provides the stability results corresponding to (27) and (28) for the buck and boost configurations.

3.2. Stability analysis

To show that the primary controller ensures stability of the equilibrium point given by (17), an eigenvalue analysis is performed. The circuit parameters used for simulations and experiments are the following: $E_1 = 36$ V, $E_2 = 28$ V, $C_1 = C_2 = 660\mu$F, $L_1 = L_2 = 400\mu$H, $R_{d1} = 0.4\,\Omega$, $R_{d2} = 0.3\,\Omega$, $V_{nl} = 32$ V, $P = 130$ W. Eigenvalues of the **A** matrix are obtained in the following cases and are shown in Fig. 5. Initially, the circuit is operated in open loop with $k_{OL,1} = 0.471$ and $k_{OL,2} = 0.533$. The eigenvalues for this case are indicated as $\boldsymbol{\lambda}_{\mathbf{OL}}$. Droop controllers may introduce low frequency oscillations around the equilibrium point [52,53]. In the case of buck-boost configuration, when the microgrid voltage is drooped in proportion to the output current of the converter, it results in unstable equilibrium point. Consider primary control inputs to the converter given by

$$d_{j,pri} = \frac{k_{OL,j}E_j - \frac{P}{x_{\mu g}}R_{dj}}{E_j},\ j = 1,\ 2 \tag{34}$$

where $x_{\mu g}$ is the instantaneous microgrid voltage. Eigenvalues calculated for this case are indicated as $\boldsymbol{\lambda}_{\mathbf{conv}}$. For the next case, consider primary controllers given by (8). To show that insufficient droop resistances may result in an unstable equilibrium point, consider the case $R_{d1} = 0.02$, $R_{d2} = 0.015$. Eigenvalues for this case are indicated as $\boldsymbol{\lambda}_{\mathbf{unstable}}$ which are $5 \pm$ j973 and $-86 +$ j0. For each of the above mentioned cases, it can be observed that there is at least one eigenvalue in the right half plane (RHP) indicating instability. Next consider increasing the droop resistances to $R_{d1} = 0.4\,\Omega$, $R_{d2} = 0.3\,\Omega$. Eigenvalues for this case are indicated as $\boldsymbol{\lambda}_{\mathbf{stable}}$. For this case, P is increased in steps from 130 W and the dominant eigenvalues are plotted. For this case, all the eigenvalues are shifted to the LHP. It can be observed from Fig. 5 that a stable equilibrium point is obtained till $P = 342$ W. To consider the effect of line impedances in droop-controlled microgrids, the original system is augmented as shown in Fig. 6. A third buck-boost converter is also added. The additional circuits in Fig. 6 represent

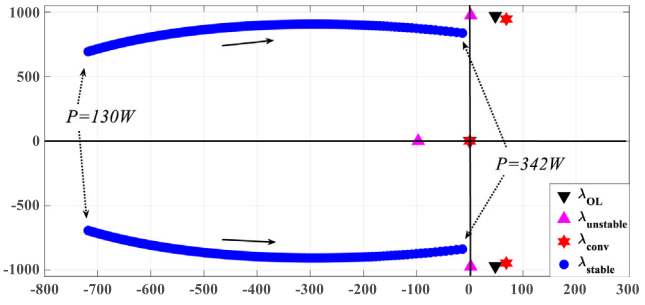


Fig. 5. Eigenvalues for DC microgrid in Fig. 1 ($m = 2$) for different test cases.

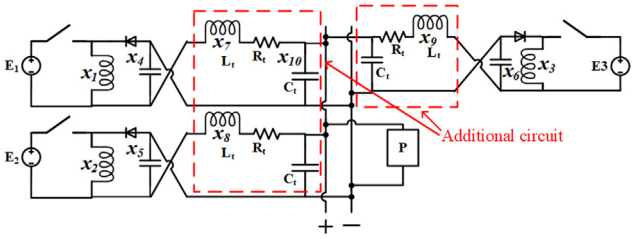


Fig. 6. Augmented microgrid with additional elements.

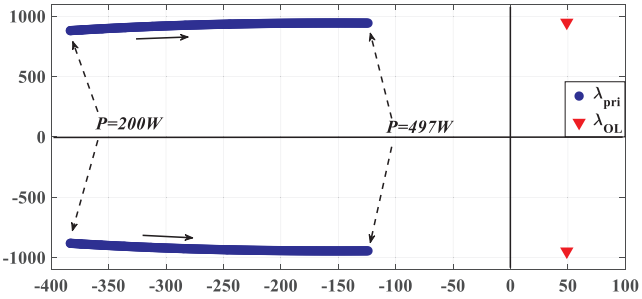


Fig. 7. Eigenvalues for the augmented microgrid in Fig. 6.

connecting lines [50]. In microgrids $X_L \ll R_L$ where R_L and X_L are the resistance and reactance of the lines [54–56]. Thus, connecting line inductances are negligible in a DC microgrid. However, in a general case, the circuit shown in Fig. 6 represents line impedance including the addition of a decoupling capacitor which are almost always added in this type of circuits. Modified circuit parameters are $E_3 = 30$ V, $k_{OL,3} = 0.516$, $R_{d1} = 0.2\,\Omega$, $R_{d2} = 0.22\,\Omega$, $R_{d3} = 0.15\,\Omega$, $L_t = 10\mu$H, $R_t = 0.21\,\Omega$, $C_t = 16.67\mu$F. Firstly eigenvalues of **A** are evaluated in open loop and those that lie on RHP are shown as $\boldsymbol{\lambda}_{\mathbf{OL}}$ in Fig. 7. For the next case, consider the primary controller given in (8). The load power P is increased in steps from 200 W. The locus of dominant eigenvalues is also plotted in Fig. 7 indicating that the proposed controller maintains the stability of the system till a maximum P of around 500 W. This stable behavior is also observed when evaluating all of the ten eigenvalues of the augmented system represented in Fig. 6. Table 4 gives dominant eigenvalues λ_d in Fig. 5 and Fig. 7 at discrete values of P. The stability analysis performed in this section is more accurate compared to the analysis performed in [57]. In [57], a system similar to Fig. 6 is considered. But, the system model takes into account the state equations of the *RLC* filter alone. In case of boost and buck-boost topologies, converter L, C values are much higher compared to L, C values of filters and connecting lines. In this paper, the CPL is realized by loading the microgrid with a buck converter acting as a POL converter, whose output voltage is maintained constant using a fast regulating PI

Table 4
Largest eigenvalues at discrete P for original and augmented microgrids. Refer Fig. 5 and Fig. 7.

	λ_d for microgrid in	
P(W)	Fig. 1($m = 2$)	Fig. 6
130	$-717.7 \pm$ j693.9	$-417.2 \pm$ j858.2
170	$-656.6 \pm$ j759.9	$-398.5 \pm$ j871.4
210	$-581.7 \pm$ j819.4	$-378 \pm$ j884.3
240	$-512.1 \pm$ j858.5	$-361.2 \pm$ j893.7
270	$-424.4 \pm$ j890	$-343 \pm$ j902.8
300	$-306.1 \pm$ j906.4	$-323.3 \pm$ j911.4
330	$-125.2 \pm$ j881.8	$-301.6 \pm$ j919.6
342	$-13.9 \pm$ j837.7	$-292.4 \pm$ j922.7
400	unstable	$-241.8 \pm$ j935.7
450	unstable	$-187.9 \pm$ j943
497	unstable	$-124 \pm$ j943.6

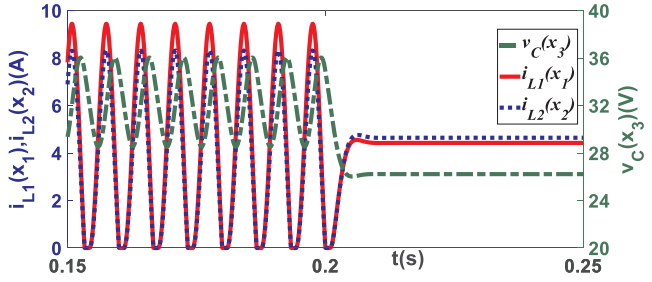


Fig. 8. Simulation results for parallel buck-boost converters transitioning from open loop to primary control.

controller [14]. The parameters of the POL converter are as follows. L_L = 300μH, C_L = 1680μF, V_L = 16.2 V, R_L = 2 Ω.

4. Simulation and Experimental results

4.1. Simulation results

Simulation and experiments based on the circuit in Fig. 1 are performed to show the validity of the analysis.

4.1.1. Case 1: With primary control only

The simulation results for transition from open loop to primary control are given in Fig. 8. For t < 0.2 s, limit cycle oscillations of around 150 Hz are observed in the current and voltage waveforms. The primary controllers (8) are switched on at t = 0.2 s and the limit cycle oscillations are damped in around 10 ms. It can be observed from Fig. 8 that i_{L1} (x_1), i_{L2} (x_2) and v_C (x_3) settle down at equilibrium values of 4.43A, 4.64A and 26.2 V respectively. The current sharing ratio, given by I_{L2eq}:I_{L1eq} (X_{2eq}:X_{1eq}) equals 1.05. The input currents of the two converters are $I_{in1,pri}$ = 1.87A and $I_{in2,pri}$ = 2.25A and are shared in the ratio $I_{in2,pri}$:$I_{in1,pri}$ = 1.20.

4.1.2. Case 2: Addition of secondary control

A secondary controller with k_i = 0.9 is added at t = 0.5 s. Fig. 9 shows the simulation results for the transition from primary to secondary control mode. It can be observed from Fig. 9 that v_C (x_3) raises to 32 V. The currents, i_{L1} (x_1) and i_{L2} (x_2) initially overshoot to around 7A individually. However, they quickly decrease and settle down at their steady state values of 4.02A and 4.16A respectively. The current sharing ratio, given by $I_{L2eq,sec}$:$I_{L1eq,sec}$ ($X_{2eq,sec}$:$X_{1eq,sec}$) equals 1.04. The corresponding input current values are $I_{in1,sec}$ = 1.89A and $I_{in2,sec}$ = 2.22A and the ratio is $I_{in2,sec}$:$I_{in1,sec}$ = 1.18.

4.2. Experimental results

Digital control implemented in the experimental setup is shown in Fig. 10. Texas Instruments TM4C123 launch-pads are used as digital controllers, which includes a communication interface using the CAN

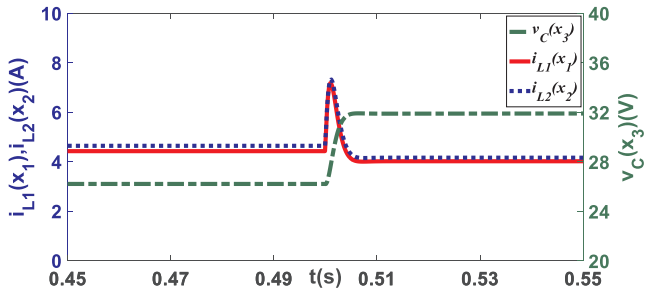


Fig. 9. Simulation results for parallel buck-boost converters transitioning from primary control to secondary control.

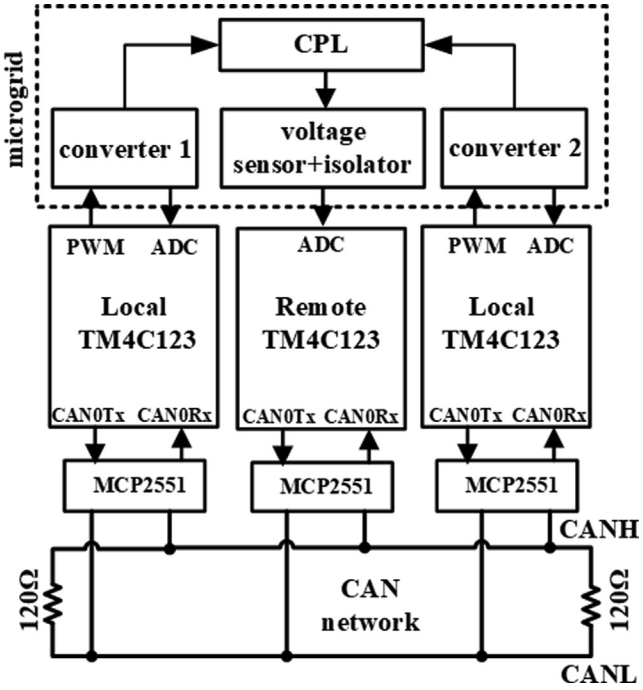


Fig. 10. Digital control and CAN network.

protocol [58]. A total of 3 launch-pads are used. Two launch-pads are used for the local control of paralleled converters. One launch-pad is used to sense the microgrid voltage and communicate secondary control signals to the paralleled converters.

4.2.1. Case 1: With primary control only

The open loop waveforms of the state variables are shown in Fig. 11. During open loop operation, x_1 and x_2 depend on internal impedance of the two input voltage sources. Transition from open loop to primary control is given in Fig. 12. A solid line is drawn in Fig. 12 which separates the current and voltage waveforms in open loop from the corresponding waveforms after primary controller is inserted. To the left of the solid line, limit cycle oscillations identical to Fig. 11 can be observed. To show that the oscillations disappear with the proposed control, the state-variable dynamics after the primary control is inserted, the timescale in Fig. 12 is increased to 50 ms/div instead of 10 ms/div in Fig. 11. Looking at Fig. 12, one can observe the following. As soon as the primary controller is switched on, limit cycle oscillations are damped in about 30 ms. The currents and voltages settle down at their equilibrium values. The primary controller enables current

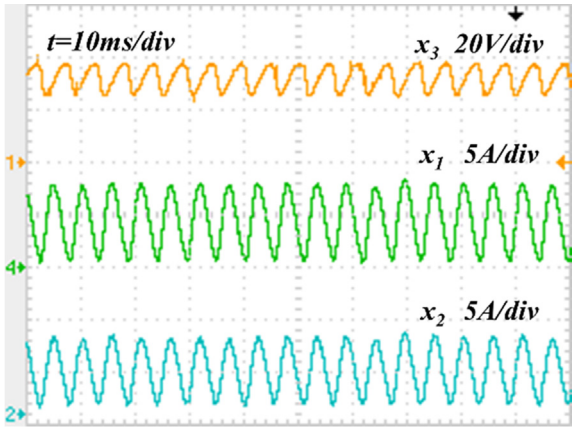


Fig. 11. Experimental results of parallel buck-boost converters-open loop.

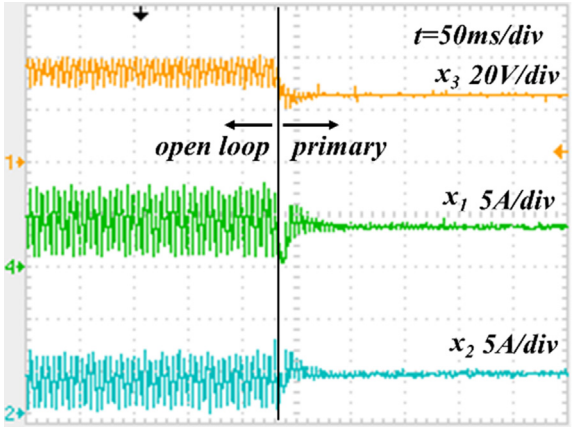


Fig. 12. Experimental results for parallel buck-boost converters transitioning from open loop to primary control.

sharing with a ratio is given by X_{2eq}:X_{1eq} = 1.05.

4.2.2. Case 2: Addition of secondary control

The results showing the transition from primary to secondary control mode is given in Fig. 13. As soon as the secondary controller is switched on, the microgrid voltage raises to 32 V. The two inductor currents x_1 and x_2 slightly decrease and settle down to their new equilibrium values in about 50 ms. The current sharing ratio is given by $X_{2eq,sec}$:$X_{1eq,sec}$ = 1.04 which is almost identical to that yielded by the primary control stage. Secondary controllers enable line and load regulation. To test line regulation, E_2 is reduced to 23 V and the results are given in Fig. 14. It can be observed that x_2 decreases to 1.88A, x_1 increases to 6.19A and x_3 is restored to 32 V within 150 ms. To test load regulation, the load power is increased from 130 W to 180 W and results are given in Fig. 15. It can be observed that x_1 and x_2 increase to their new equilibrium values of $X_{1eq,sec}$ = 5.55A, $X_{2eq,sec}$ = 5.75A. The microgrid voltage, $X_{3eq,sec}$ is restored to 32 V within 70 ms. The sharing of inductor currents, $X_{2eq,sec}$:$X_{1eq,sec}$ = 1.04 and the input currents, $I_{in2,sec}$:$I_{in1,sec}$ = 1.17 are identical to those obtained in the primary and secondary control stages.

5. Conclusions and future directions

A control scheme for parallel connected DC-DC converters is developed for DC microgrids loaded with CPLs. The control scheme consists of two levels. The lower control level, namely the primary control, enables current sharing and also damps limit cycle oscillations due to the CPL. To compensate for the voltage deviations due to the primary

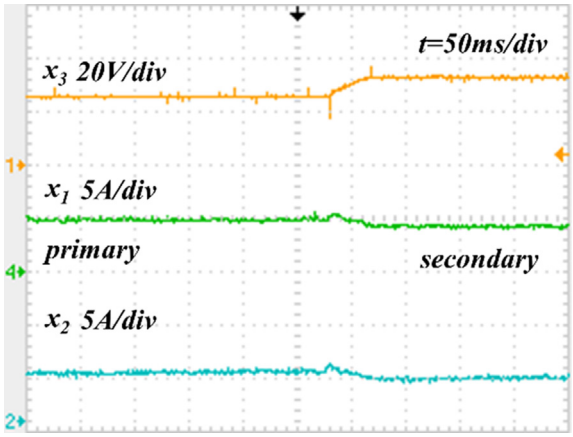


Fig. 13. Experimental results for parallel buck-boost converters transitioning from primary to secondary control.

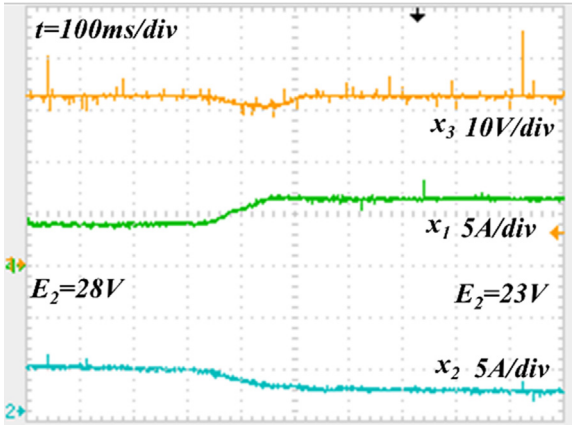


Fig. 14. Experimental results for line regulation of parallel buck-boost converters.

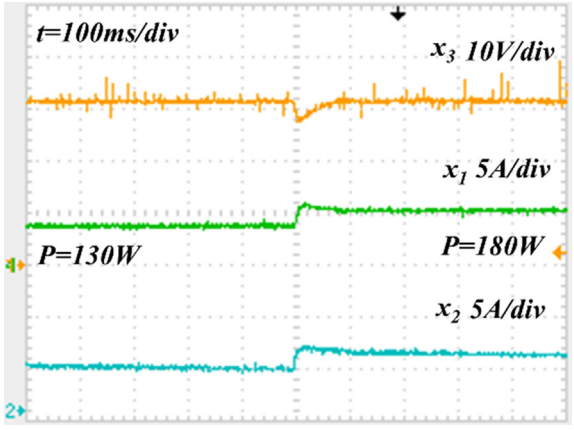


Fig. 15. Experimental results for load regulation of parallel buck-boost converters.

controller, secondary control is added. The secondary control communicates to primary controllers using a high speed communication link based on a CAN protocol. The secondary control gains are designed to ensure that current sharing obtained using the primary controller is maintained. The stability conditions are derived and explained using the equivalent circuit of constituent converters. The robustness and scalability of the control scheme is also demonstrated. The results obtained for three basic topologies namely buck, boost and buck-boost are compared. To enable interfacing of sources with a wide voltage range, a prototype microgrid containing parallel buck-boost converters is considered. Simulation and experimental results in such a microgrid are provided to verify the analysis.

The concept of multiple microgrid clusters is gaining interest from researchers. Hence, a natural next step for this research is to explore such concept and develop a voltage stability index for the individual microgrids in a cluster. Such an index could determine the degree of influence each microgrid wields on the stability of the cluster as a whole.

CRediT authorship contribution statement

Mahesh Srinivasan: Conceptualization, Formal analysis, Software, Validation, Investigation, Writing - original draft. **Alexis Kwasinski:** Writing - review & editing, Supervision, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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